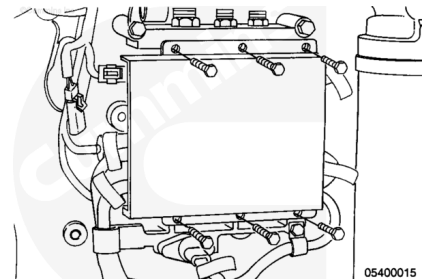


Trainings ▾

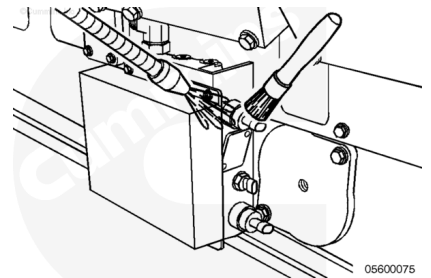
Remove

Remove the engine control manual (ECM). See the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113.



⚠ WARNING ⚠

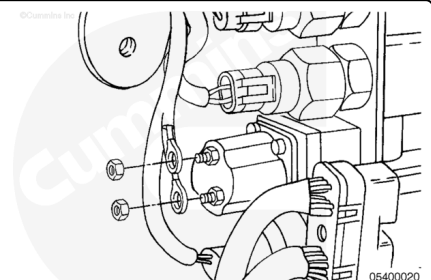
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



Clean the fuel shutoff valve and surrounding area. Use solvent.

Remove the nut holding the electrical connection of the fuel shutoff valve coil. Use a 3/8 inch wrench.

Remove the connection.



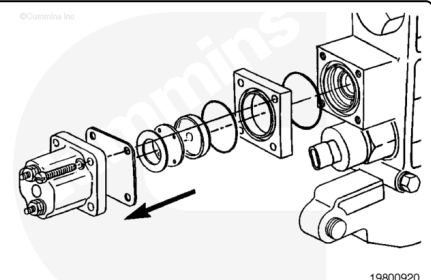
Remove the four capscrews.

Remove the coil housing and the fuel shield.

Discard the o-ring.

Remove the spring washer, valve disc, actuator disc, and actuator spacer from the valve housing.

Discard the o-ring.

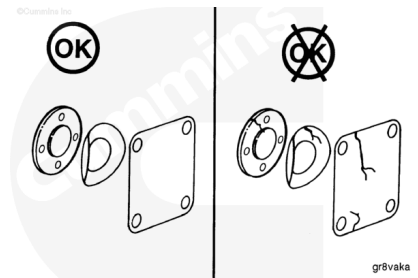


Inspect for Reuse

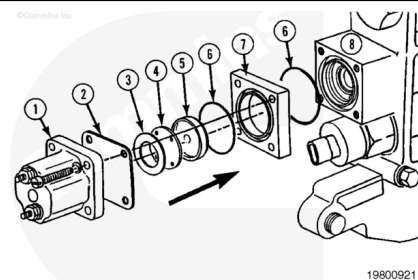
Clean all the parts **except** the coil assembly. Use mineral spirits.

Note : Do not get solvent on the coil. Clean the coil with a dry cloth. Use a 200 grit emery cloth and a flat surface to polish the coil surface.

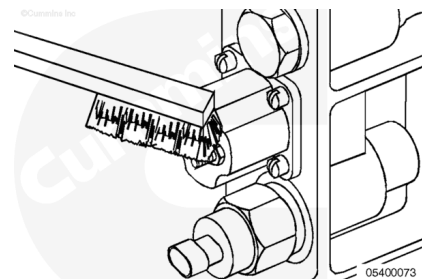
Inspect the valve disc, valve seat, and actuator disc for dirt, metal parts, bonding separation, corrosion, cracks, or wear. Replace if necessary.



Inspect the fuel shutoff valve housing (7 and 8) for wear or other damage.



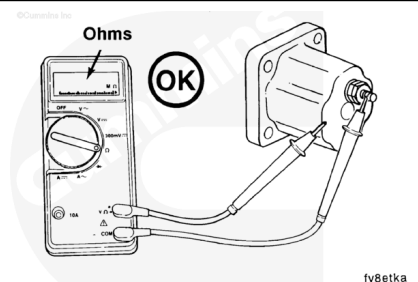
Clean any corrosion from the coil terminal. Use a wire brush.



Measure the resistance of the coil with a multimeter.

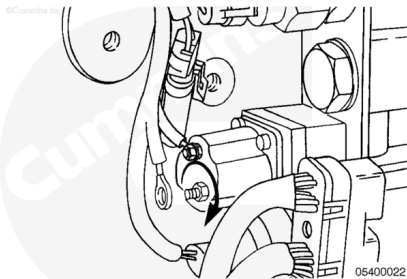
Replace the coil if it does **not** have 28 to 32 ohms of resistance.

Note : If the coil assembly shows 0 ohms, there is an electrical short in the coil.



Tighten the nut that holds the electrical connection post on the fuel shutoff valve coil.

Torque Value: 3 n•m [27 in-lb]



Install

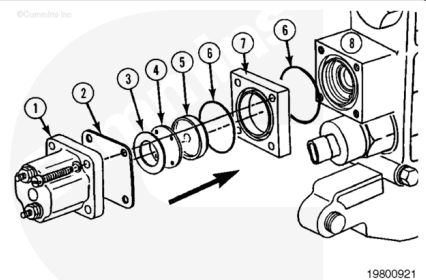
Assemble the shutoff valve as shown. Install these parts as follows.

Install a new o-ring (6) between the spacer (7) and the electronic control valve body (8).

Install the spacer (7) o-ring groove toward the coil.

Install the actuator disc (5) with the cup side toward the coil.

Install the spring washer (3) with the cup side toward the coil.



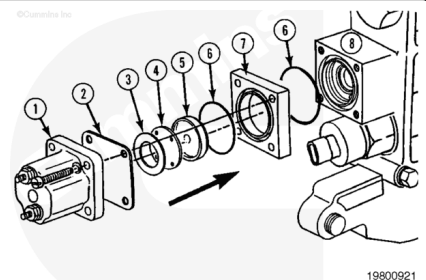
Align the actuator disc (5), spacer (7), and valve disc (4) on the electronic control valve body (8).

Install new o-rings (6) on the spacer (7).

Note : The coil must be orientated with the electrical connection post on the bottom.

Note : Make sure the shutoff valve coil is the correct voltage (24 VDC). The coil voltage and part number are cast into the terminal connection end of the coil.

Install the fuel shield (2) and coil (1) on the front cover (8).



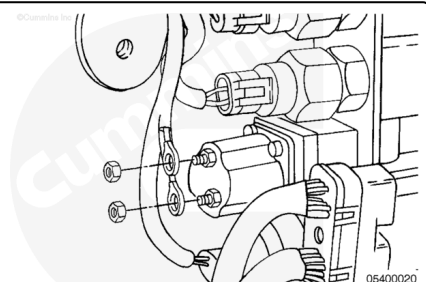
Torque Value: 8 n•m [71 in-lb]

Install the electrical connection on the fuel shutoff valve.

Install the nut on the threaded post of the coil.

Hold the post of the nut firmly while tightening the connection nut. Use two 3/8 inch wrenches.

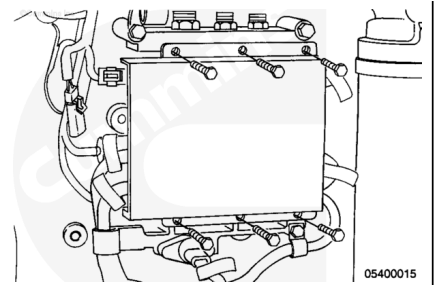
Torque Value: 2 n•m [18 in-lb]



Install the ECM to the control valve body. See the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113.

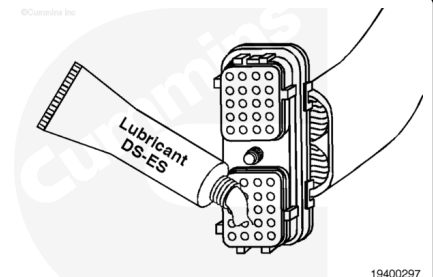
Tighten the six ECM mounting capscrews.

Torque Value: 8 n•m [71 in-lb]



⚠ CAUTION ⚠

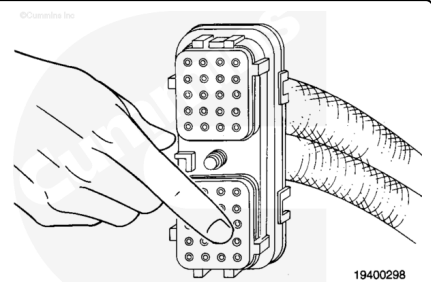
Use only Cummins® recommended lubricant DS-ES, Part Number 3822934. Other lubricating oil or grease in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.



Apply a thin coat of lubricant to the connector nosepiece.

Spread the lubricant across the connector nosepiece so it enters every pin hole and lubricates the contacts.

Lubricant **must not** be visible on the surface of the nosepiece.

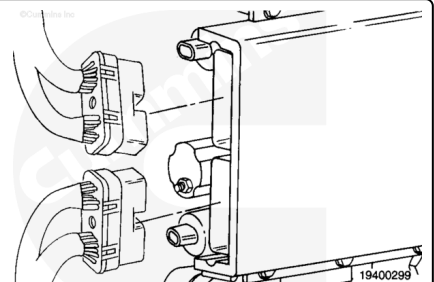


Connect the original equipment manufacturer and engine harness connectors to the ECM.

Tighten the connector capscrews.

Torque Value: 3 n•m [27 in-lb]

Note : When an ECM is replaced, the new ECM must be calibrated. See the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113.



Voltage Check

Use a multimeter to check the voltage to the coil. Measure the voltage from the solenoid's supply connection to the engine block ground. The multimeter **must** show battery voltage.

Crank the engine to provide voltage to the fuel shutoff valve terminals.

Once the ECM has received the 50-rpm signal, the voltage will remain supplied to the fuel shutoff valve until the keyswitch is cycled to the OFF position.

